Register Number: 192425119

Question 2: Weather and Accident Severity

Research Question

Is there an association between weather conditions and accident severity?

Hypotheses

- H0: Weather condition and accident severity are independent.
- H1: Weather condition and accident severity are associated.
- Significance level $\alpha = 0.05$

Formula

$\chi^2 = \Sigma\ (O - E)^2 / E$

$E = (\text{Row total} \times \text{Column total}) / \text{Grand total}$

$df = (\text{rows} - 1)(\text{columns} - 1) = 3 \times 2 = 6$

Data Used (read from reference figure)

Weather	Low	Medium	High	Total
Clear	20	16	6	42
Rainy	8	13	9	30
Foggy	3	6	6	15
Cloudy	7	4	2	13
Total	38	39	23	100

Calculation

- Expected counts (E) are computed from the row and column totals above.
- Sum of $(O - E)^2 / E$ across all 12 cells gives $\chi^2 \approx 8.80$, df = 6.

Result

Quantity	Value
χ^2 statistic	8.80
Degrees of freedom	6
p-value	0.185

Conclusion

The p-value (0.185) is greater than 0.05, so we fail to reject H0. The data do not show a statistically significant association between weather condition and accident severity.

Interpretation Notes

- Some expected counts are small (for example Foggy/High ≈ 3.5 and Cloudy/High ≈ 3.0). The chi-square approximation is less reliable under those conditions; Fisher's exact test (fisher.test in R) is a useful check.
- Visually, Rainy weather shows a higher share of Medium and High severity accidents than Clear weather, but the difference is not large enough to be significant with these counts.

Graph

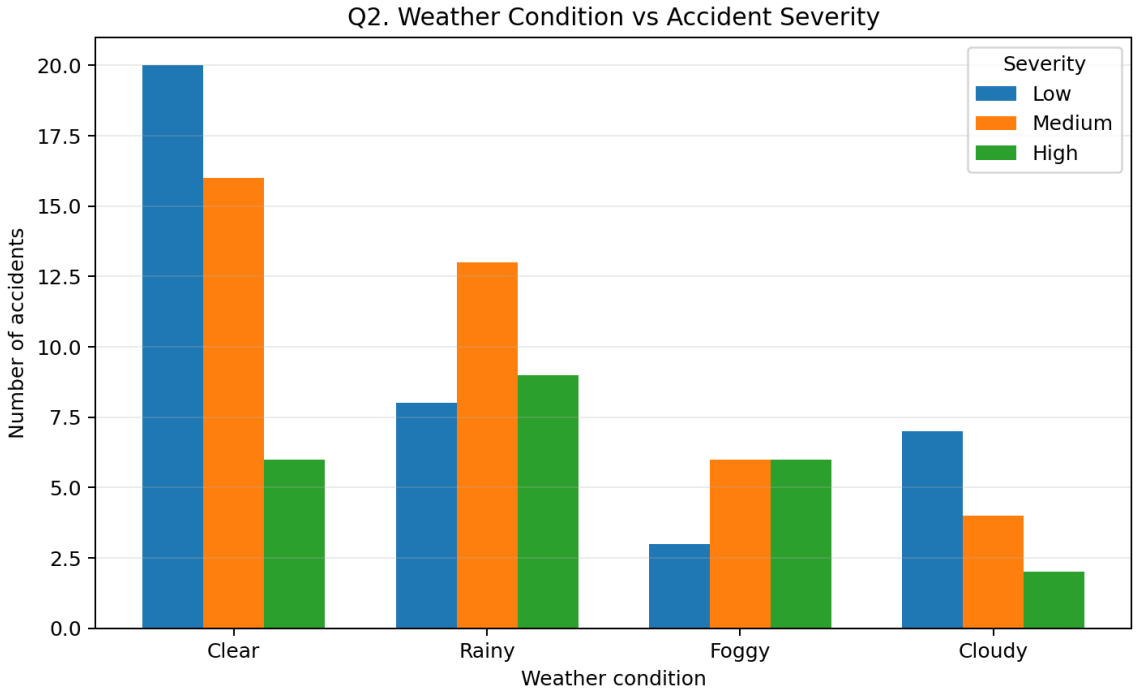


Figure 2. Reference graph for Question 2.

Note: the counts and values in this document were read from the reference figure. Re-run the test on the original project dataset in R (e.g. chisq.test, t.test, aov, cor.test) to confirm the final numbers before submission.